Project Design Phase-II

Data Flow Diagram & User Stories

Date	27 October 2026
Team ID	SWUID2026403463
Project Name	HotelPro – Hotel Management System
Maximum Marks	4 Marks

Data Flow Diagrams:

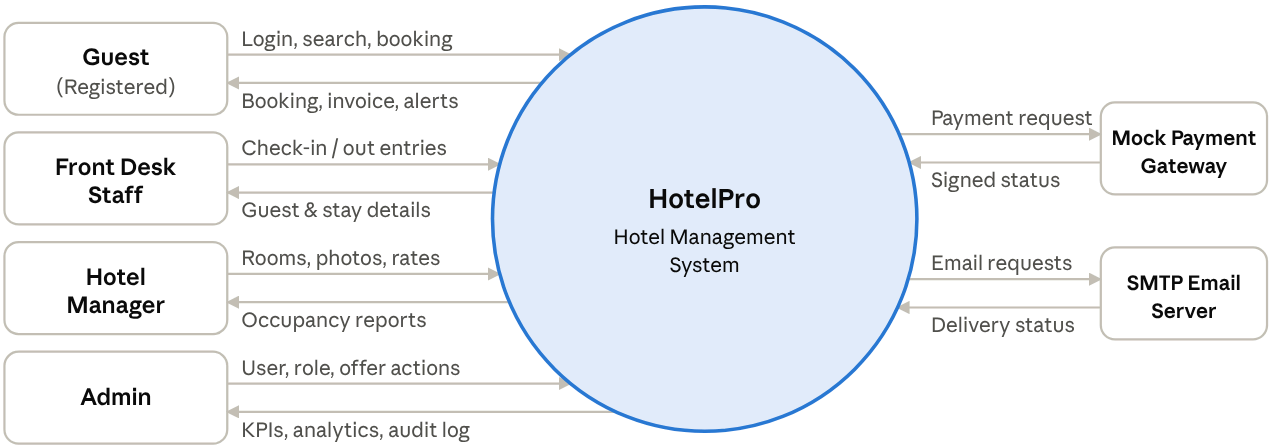
A Data Flow Diagram (DFD) illustrates how data moves within the HotelPro hotel management platform. It shows how users (guests, front desk staff, hotel managers and admins) and external systems (mock payment gateway and SMTP email server) interact with the system, how information flows between processes, and where the data is stored. The diagrams below follow the project's functional specification and are drawn at two levels: Level 0 (context) and Level 1 (major processes).

Legend:

Symbol	Meaning
Rectangle	External entity (user or external system)
Circle / rounded box	Process that transforms data
Open-ended bar (D1–D10)	Data store (database collection group)
Arrow	Data flow, labelled with the data moved

DFD Level 0 – Context Diagram:

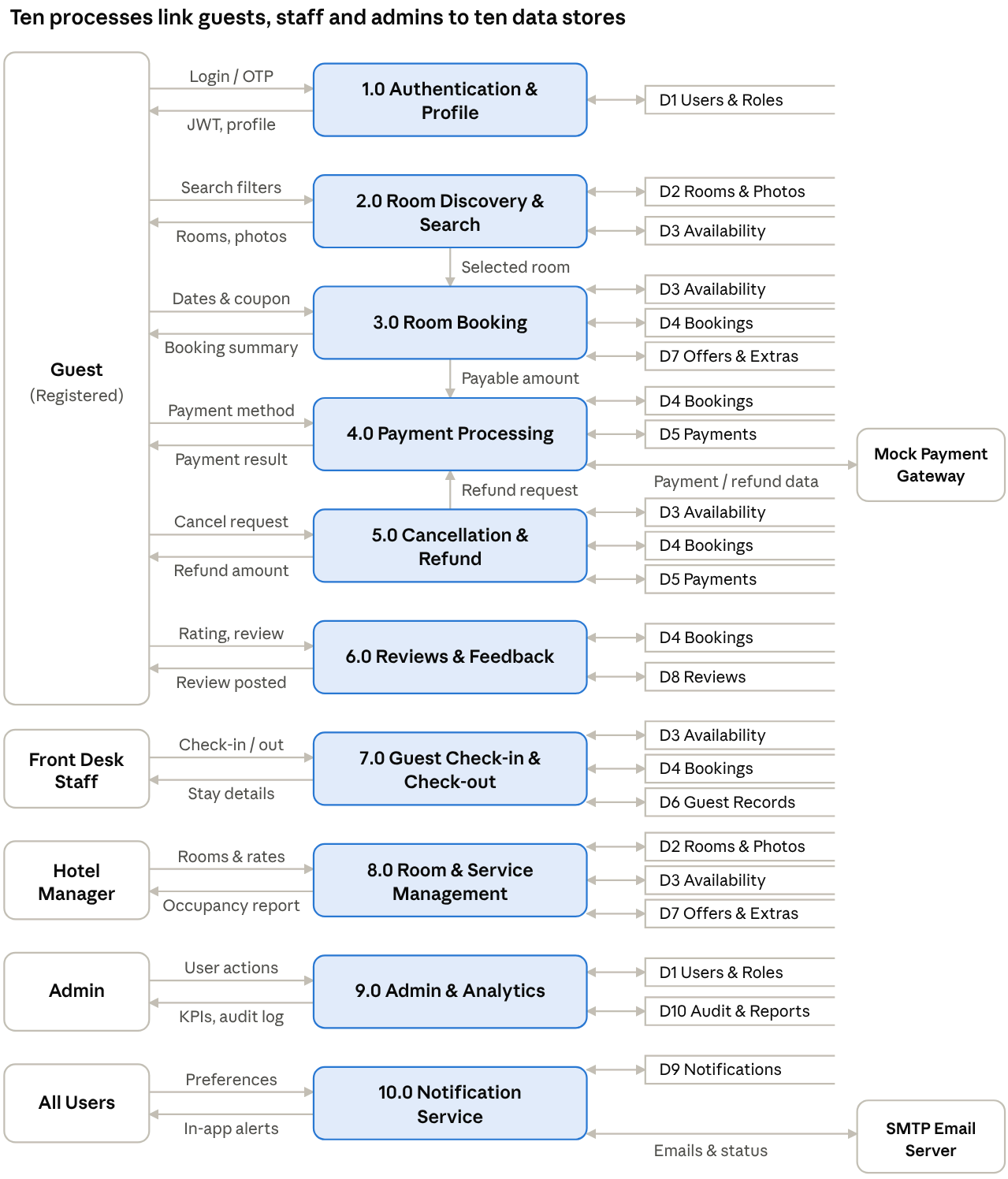
HotelPro exchanges data with four user roles and two external systems



context diagram · 4 user roles, 2 external systems

Figure 1: DFD Level 0 – HotelPro

DFD Level 1 – Major Processes:



Shaded boxes are processes; double-headed arrows on the right show reads and writes on data stores.

level 1 diagram · 10 processes, 10 data stores

Figure 2: DFD Level 1 – HotelPro (processes 1.0 to 10.0, data stores D1 to D10)

Level 1 Process Description:

No.	Process	Description	Data Stores
1.0	Authentication & Profile	Registration, OTP/email verification, separate guest and staff/admin login, JWT issue, password reset, profile and RBAC	D1
2.0	Room Discovery & Search	Room types, photos, amenities and rates; search by dates, guests and room type; filter and sort	D2, D3
3.0	Room Booking	Availability check, temporary room hold with 10-minute expiry, booking summary with services, coupon, taxes and fees; booking status handling	D3, D4, D7
4.0	Payment Processing	Mock gateway initiate, verify, refund; signature check; server-side amount; invoice generation	D4, D5
5.0	Cancellation & Refund	Cancellation before the deadline, refund slabs, room release, cancellation confirmation	D3, D4, D5
6.0	Reviews & Feedback	Ratings and reviews for completed stays, average rating update	D4, D8
7.0	Guest Check-in & Check-out	Guest records, ID details, check-in, check-out, final bill and stay history	D3, D4, D6
8.0	Room & Service Management	Rooms, room photos stored in the database, rates, services, offers and coupons	D2, D3, D7
9.0	Admin & Analytics	KPIs, user and role management, occupancy and revenue reports, audit log	D1, D10
10.0	Notification Service	In-app notifications and asynchronous emails with retry	D9

User Stories

User Story Table – HotelPro

User Type	Functional Requirement (Epic)	User Story / Task	Acceptance Criteria	Priority	Release
Guest	Authentication	As a guest, I can register and verify my account by email or OTP so that I can book rooms.	Account is created; unverified users cannot book.	High	Sprint-1
Guest	Room Discovery	As a guest, I can search rooms by dates, guests and room type, and filter by price and amenities.	Results match the dates, capacity, room type and price filters; room photos are shown.	High	Sprint-1
Guest	Room Booking	As a guest, I can select a room and hold it temporarily while I complete my booking.	Room is held for 10 minutes with a countdown; no double booking.	High	Sprint-2
Guest	Booking Summary	As a guest, I can add services and a coupon and see a full booking summary.	Total shows room charges, taxes, services, discount and grand total.	High	Sprint-2
Guest	Payment	As a guest, I can pay using the mock gateway (UPI, card, net banking, wallet).	Success confirms the booking; failure releases the room.	High	Sprint-2
Guest	Booking Confirmation	As a guest, I can get a booking confirmation with an invoice and download it as PDF.	Confirmation appears in My Bookings and is emailed.	High	Sprint-2
Guest	Cancellation	As a guest, I can cancel a booking before the deadline and see my refund.	Refund breakup is shown; booking becomes Cancelled then Refunded.	Medium	Sprint-3
Hotel Manager	Room Management	As a hotel manager, I can add rooms with photos and set room rates and services.	Rooms appear to guests with their photos; photos are stored in the database.	High	Sprint-1

User Type	Functional Requirement (Epic)	User Story / Task	Acceptance Criteria	Priority	Release
Front Desk Staff	Check-in / Check-out	As front desk staff, I can check a guest in and out and record guest details.	Check-in is allowed only for a confirmed booking; room status updates after check-out.	Medium	Sprint-3
Hotel Manager	Reports	As a hotel manager, I can view occupancy and revenue reports for my hotel.	Report matches bookings and payments for the selected period.	Medium	Sprint-3
Guest	Reviews	As a guest, I can rate and review a hotel stay I completed.	Review appears with updated average rating.	Medium	Sprint-3
Admin	Administration	As an admin, I can manage users, staff accounts and offers, and view analytics and the audit log.	Actions are logged; dashboard shows KPIs and charts.	High	Sprint-3